Analysis of Crop-Specific Agriculture to Fire Hazard Severity Zones in California

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Introduction

Over the past few years, the impact of wildfire damages have grown substantially across California, becoming increasingly frequent, extensive, and severe. According to long-term wildfire tracking data, all but two of the 20 largest wildfires in state history have occurred since 2000, with several of the top fires burning hundreds of thousands to over a million acres of California in recent years[1]. Multiple megafires in 2020 scorched more than 4.3 million acres, while 2024 saw a total of 8,024 wildfire incidents that burned over 1,050,000 acres statewide. These past incidents demonstrate the threat escalation of wildfires in California, as the number of cases and damages far exceeds historical averages, underscoring the growing challenges for communities, ecosystems, and economic sectors across the state. Thus, studies on the impact and potential regional risk of wildfires have grown more prominent, with a focus on regional susceptibility through combinations of dry vegetation, wildland-urban interface development, and climate-driven weather conditions[2].

On the other hand, a large portion of California's landscape has been dedicated to agricultural use. California's agricultural sector consistently ranks as the most valuable in the United States, with farms and ranches generating around $61.2 billion in cash receipts in 2024, producing more than 400 distinct commodities[3]. California's agriculture supplies over one-third of the nation's vegetables and more than three-quarters of its fruits and nuts. According to the California Department of Food and Agriculture, top commodities in terms of cash receipts include dairy products, grapes, cattle and calves, lettuce, almonds, and strawberries, underscoring the economic diversity of California agriculture. Furthermore, California's agriculture also plays a critical role in the export markets, with billions of dollars in agricultural exports driven by tree nuts, dairy, wine, and other staples[4].

As a result, it is important to emphasize how California's agricultural production could potentially be impacted by the increasing wildfire activities. Previous research has already documented the impacts of wildfires on agricultural lands, illustrating how fire events disrupt production, damage infrastructure, and impose economic and health burdens on farming

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[1] CalMatters. (2025). *Track California wildfires* [Interactive wildfire map]. CalMatters. https://calmatters.org/california-wildfire-map-tracker/

[2] Panish | Shea | Ravipudi LLP. (2025, May 29). *Top 5 California counties most susceptible to wildfires in 2026*. Panish Law. https://www.panish.law/2025/05/top-5-california-counties-most-susceptible-to-wildfires/

[3] California Department of Food and Agriculture. (n.d.). *California agricultural production statistics*. CDFA. https://www.cdfa.ca.gov/Statistics/

[4] Dutton, J. (2025, May 9). *How agriculture makes California a leader in the global economy*. *The Packer*. https://www.thepacker.com/news/industry/how-agriculture-makes-c alifornia-leader-global-economy

communities. For example, recent studies highlight that wildfire exposure leads to crop losses, smoke contamination, market closures, and operational disruptions for producers across California, with effects ranging from short-term financial shocks to long-lasting structural damage[5]. Other research on wildfire crisis responses, such as evaluations of Agricultural Pass programs, emphasizes how wildfire events affect readiness infrastructure and broader agricultural resilience, highlighting that farms and ranches are integral to emergency frameworks beyond urban settings[6]. However, there are few findings on existing wildfire research and studies that have quantified California's agricultural exposure to fire hazard at a spatial landscape scale, nor are there widely available statistical risk assessments tied directly to commodity-specific vulnerability. Many studies focus on general ecological and economic impacts, emergency responses, or urban and forest risk patterns, but omit detailed assessments of how specific crops or agricultural regions intersect with regions that are prone to wildfires. This gap leaves a limited understanding of which crops and agricultural areas face the greatest exposure and how fire risk translates into tangible risks for production systems. Moreover, prior work has often emphasized broad environmental drivers or socio-economic effects without deeply examining crop-level spatial exposure, making it difficult to prioritize risk mitigation for high-value commodities integral to California's agricultural economy. Thus, there is a need for comprehensive, spatially explicit analyses that quantify wildfire exposure across California's agricultural landscape and assess the relative risk to different crop types.

Methodology

The study aims to fill the gap in the lack of understanding of crop-specific assessment towards wildfire vulnerability by combining multiple data sources to produce the final output. The US county-border layer was derived from GADM v4.1 (Global Administrative Areas) for the United States at Administrative Level 2 (county-level boundaries), downloaded as ESRI Shapefile format. Only features within California were selected to narrow the focus parameter. All spatial analyses were conducted using QGIS 3.44, while statistical calculations of proximity were measured in R.

Wildfire hazard data were obtained from CAL FIRE's Fire Hazard Severity Zones (FHSZ). The original files were divided into two management designations: Local Responsibility Areas (LRA) and State Responsibility Areas (SRA), with each containing polygon features classifying regions into *Moderate (1)*, *High (2)*, and *Very High* (3) hazard zones. Both LRA and SRA layers were merged into a single statewide hazard map to ensure full coverage of

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[5] Pinzón, N., Galt, R., Roche, L., Schohr, T., Shobe, B., Koundinya, V., Brimm, K., & Powell, J. (2025). *Farming and ranching through wildfire: Producers' critical role in fire risk management and emergency response* (CalAg_fire.pdf). University of California, Davis, Rangelands Center. https://rangelands.ucdavis.edu/sites/g/files/dgvnsk13956/files/media/documents/CalAg_fire.pdf

[6] Craig-Scheckman, M., Matharage, S. S., Libershal, H., Khan, M., Schmitz, G., & Aldrich, D. P. (2025). *Beyond the evacuation zone: Agricultural passes and the wildfire crisis in California*. *Natural Hazards Research*. Advance online publication. https://doi.org/10.1016/j.nhres.2025.09.007

California's landscape. Geometries were checked and corrected to ensure validity, and a vector overlay (Difference) with California's administrative boundaries (CA_admin_2) was used to identify areas outside of designated FHSZ zones. Using the Field Calculator, a new attribute was created with the expression:

```
CASE
  WHEN "FHSZ" < 0 OR "FHSZ" IS NULL THEN -3
  ELSE "FHSZI"
END
```

To assign a value of **–3** to non-wildland areas and Nulls, while preserving the original FHSZ values for wildfire-relevant areas. The final FHSZ vector layer resulted in 28233 feature entries with corresponding FHSZ codes, polygon shape length, and polygon shape area.

Agricultural land-use data were obtained from the 2024 USDA National Agricultural Statistics Service (NASS). The Cropland Data Layer (CDL) encodes overall plantation types with each assigned a unique integer class value. The file was downloaded as a GeoTIFF raster file with a native resolution of 30 meters as the unit of analysis, and the raster layer was clipped to the California admin 2 layer. Agriculture-specific types were filtered out from the overall range of plantations within the data, and the final raster identified 80 unique crop types specific to California's agriculture.

All layers were preprojected from the default EPSG:4326 - WGS 84 to EPSG:3310 NAD83 / California Albers for region-specific standardization. The FHSZ layer was rasterized to prepare for pixel-wise calculations with the CDL layer, and was divided into binary mask layers for each FHSZ category to create different levels of proximity calculations for the crop types, using the Raster Calculator Tool expression:

```
(A==1,2,3)
```

Distance to hazard zones was then quantified using the Proximity (Raster Distance) tool in QGIS, producing continuous Euclidean distance rasters for each FHSZ category. Distance values were measured in meters and preserved NoData regions outside the state boundary, and the proximity layers were each limited down to California's boundaries again using the binary mask layers' parameters with the Raster Calculator expression:

```
where(isnan(A), nan, B)
```

Setting A = the mask layers and B = the proximity layers.

To restrict calculations to agricultural areas, a binary mask raster was also created from the CDL using the Raster Calculator, where agricultural pixels were assigned a value of 1 and all

other pixels 0, with NoData preserved. The agriculture mask was then combined with the distance rasters in further raster calculations. The final agricultural vulnerability surfaces were generated using a distance-weighted raster expression in the Raster Calculator, applying categorical distance thresholds to assign higher vulnerability scores to agriculture closer to fire hazard zones. Specifically, the expression:

```
A * ((B <= 500) * 3 + ((B > 500) & (B <= 1500)) * 2 + (B > 1500) * 1)
```

Setting A = binary agriculture mask (1 = agriculture, 0 = non-agriculture) and B = proximity/distance layers for each FHSZ category (Moderate, High, Very High)

Pixels within 500 m of a hazard zone were weighted highest, 500–1500 m intermediate, and >1500 m lowest. The resulting heatmaps depict spatial variation in wildfire vulnerability across California's agricultural landscape.

To quantify agricultural vulnerability to wildfire hazard zones, all analyses were conducted in R 4.4.0 using the terra, dplyr, and purrr packages. The workflow integrated rasterized distance surfaces representing *Moderate*, *High*, and *Very High* FHSZ regions with the CDL layer for California. Each distance raster was read into R as a SpatRaster, along with the CDL raster, and a per-pixel area (in hectares) was calculated based on the CDL resolution.

A vectorized zonal statistics function was developed to extract summary statistics of each distance raster by crop type. Within the function, *NA* values in the CDL raster were replaced with zeros to ensure valid computations. For each crop zone, the function computed the mean, sum, minimum, and maximum distances to each hazard category. Pixel counts were then converted to area, and a risk score was calculated for each crop type as the ratio of the area to mean distance, providing a standardized metric of wildfire exposure. Zones with zero or no data were excluded from the analysis. The category-specific risk score was calculated as the ratio of crop area to mean distance to produce higher scores for crops that occupy larger areas and are closer to the hazard, reflecting greater exposure:

$$\text{RiskScore}_{\text{category}} = \frac{\text{Area (ha)}}{\text{Mean Distance (m)}}$$

After each categorical risk scores were calculated, the total risk score was created to combine the categorical risk scores, allowing comparison of crops by overall vulnerability to wildfire hazards, integrating exposure across all FHSZ levels:

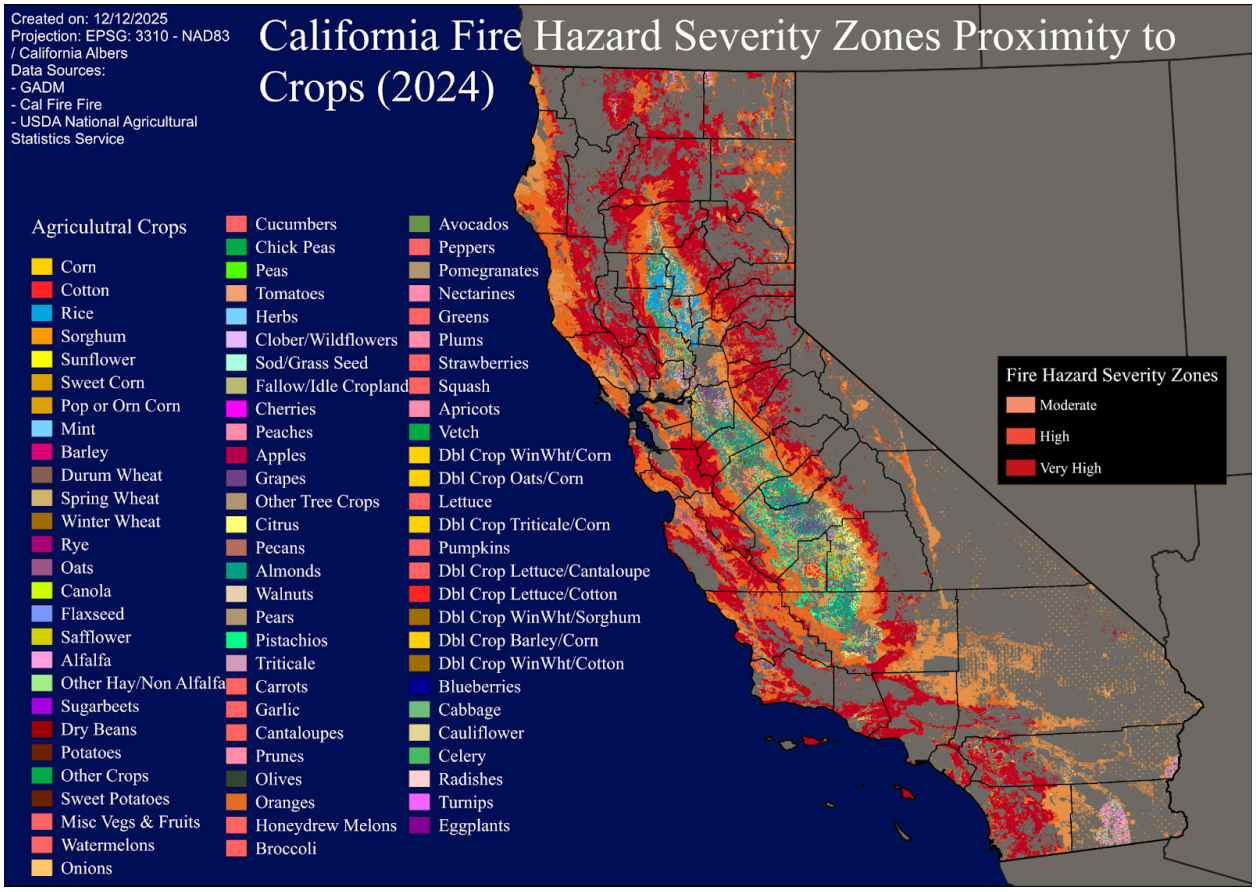
$$\text{TotalRiskScore} = \text{RiskScore}_{\text{Moderate}} + \text{RiskScore}_{\text{High}} + \text{RiskScore}_{\text{Very High}}$$

The results were merged into a single table keyed by crop code. The full results table containing per-crop RiskScores for all three hazard categories was saved as a separate CSV file. A top-ten ranking was generated by summing RiskScores across all FHSZ categories and also by individual Very High hazard exposure. This enabled the identification of both overall and category-specific high-risk crops, providing a summary of the spatial distribution of vulnerability across California's agricultural landscape.

Results

The final maps created from this study highlight agricultural crops' proximity to regions that are more prone to future wildfires. The main overlapping map demonstrates the sum of all the crops in California while integrating Cal Fire's FHSZ regions, showcasing the general locations that are potentially most at risk of being impacted by future wildfires. Furthermore, the generated heatmaps show more details on the specific locations in California that are most prone to wildfires through quantitative calculations. The maps show that regions in California's Central Valley Foothills, Northern California Interior Valleys, and Southern California Inland Valleys contain large clusters of areas that show significant exposure of agricultural lands to wildfire-prone areas.

Figure 1.



This map shows the combined layers of the Fire Hazard Severity Layer (FHSZ) and the Cropland Data Layer (CDL) to showcase the visual representation of California's agricultural proximity to wildfire-prone regions by specific crop types.
The heatmaps for each FHSZ category (Moderate, High, Very High) show agricultural areas' relative vulnerability to wildfire based on proximity to that hazard level. Values are weighted from 1 to 3, with 3 representing crops within 500 meters of the hazard (highest risk), 2 for 500–1500 meters (medium risk), and 1 for farther than 1500 meters (low risk).

Figure 2.1

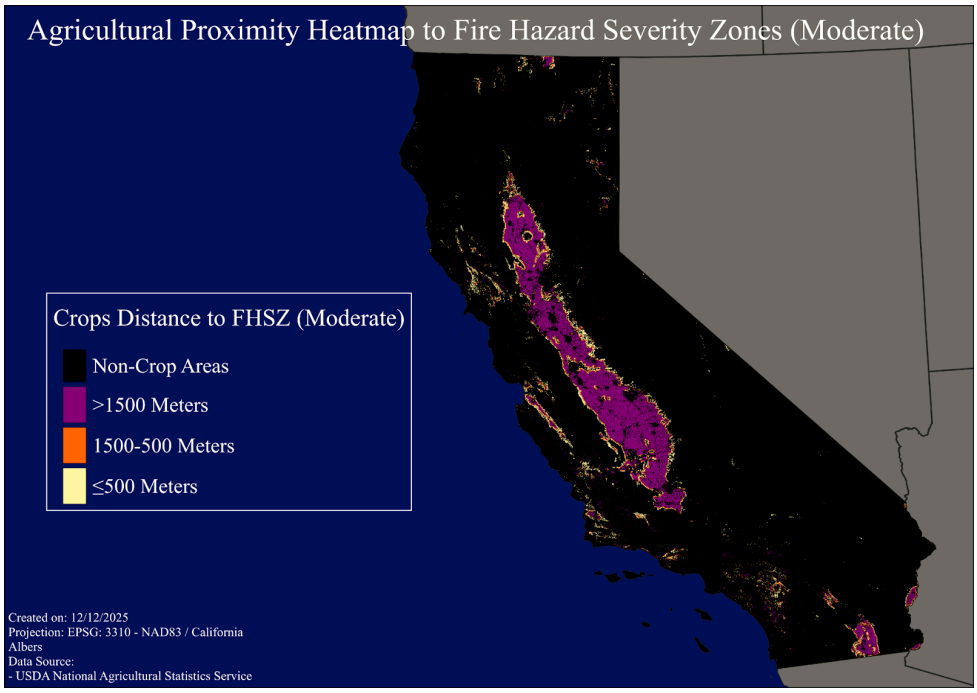


Figure 2.2

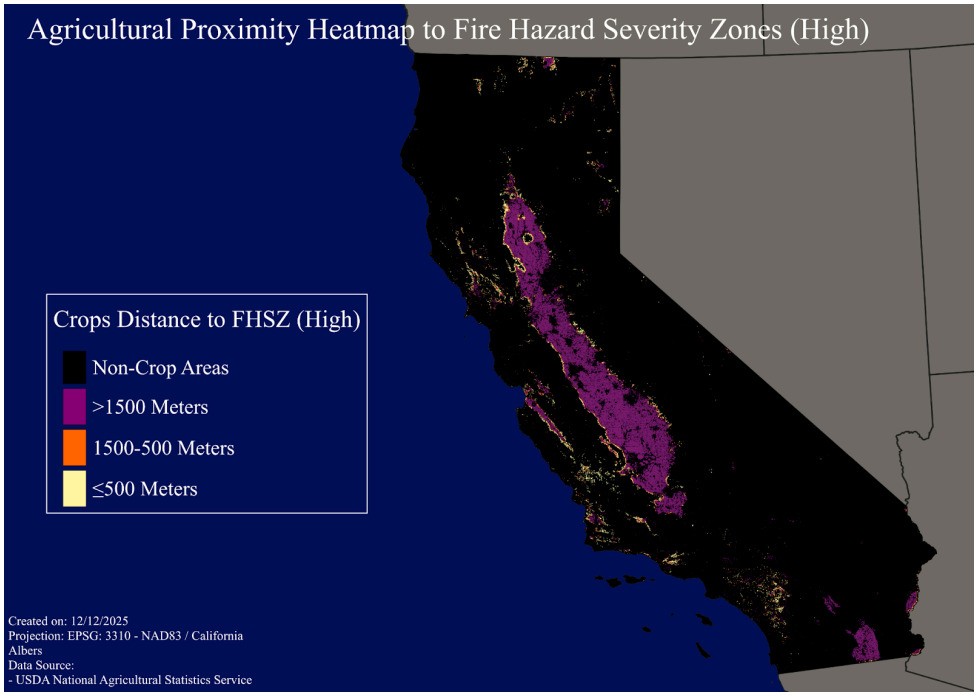
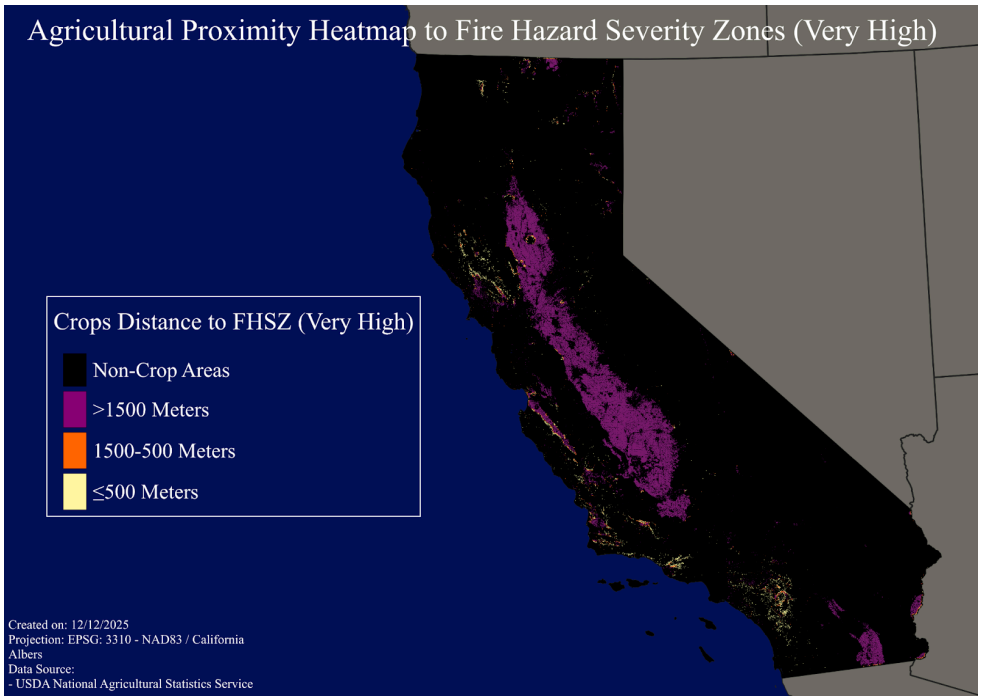


Figure 2.3



The statistical analysis of crop vulnerability to fire hazard severity zones reveals significant rankings in total risk across crop types. Almonds emerge as the most at-risk crop, with a total risk score of 184.61, primarily caused by high exposure in moderate-risk zones (96.02), high-risk (54.99), and very high-risk areas (33.60). Grapes and miscellaneous vegetables

and fruits follow suit, with overall risk scores being 103.47 and 102.51. Avocados, by contrast, exhibit the lowest total risk among perennial crops (39.88), with risk more evenly distributed across high and very high zones, indicating localized exposure. Other key crops, including rice (61.87), citrus (84.01), alfalfa (68.69), pistachios (59.67), fallow or idle cropland (51.74), and walnuts (50.10), display intermediate levels of risk. Notably, perennial tree crops such as almonds, grapes, and walnuts tend to show higher moderate-risk values, highlighting that large-scale orchards may disproportionately occupy areas with moderate fire hazard, whereas certain crops like avocados and pistachios have relatively higher proportions of risk in severe zones, potentially indicating more concentrated high-risk locations. Overall, the table shows that these crops, especially for almonds, grapes, and miscellaneous veggies & fruits, are the most at risk from future wildfire damages and should be a priority for fire mitigation interventions due to their cumulative exposure.

Crop	**Crop Code**	**"Moderate" Risk Score**	**"High Risk" Score**	**"Very High" Risk Score**	**Total Risk**
Almonds	75	96.0231215	54.98518639	33.59731096	184.6056189
Grapes	69	50.71182751	32.02961256	20.7312365	103.4726766
Misc Vegs & Fruits	47	61.58803561	24.32391413	16.59574725	102.507697
Avocados	215	9.717576049	14.02414376	16.14164669	39.88336649
Rice	3	28.60208987	20.33966644	12.92452807	61.86628438
Citrus	72	48.03623909	24.4602482	11.51059988	84.00708717
Alfalfa	36	43.45381507	14.28398294	10.95579872	68.69359673
Pistachios	204	32.46352695	16.90473597	10.30306981	59.67133272
Fallow/Idle Cropland	61	25.45670451	16.00797635	10.27174773	51.73642859
Walnuts	76	24.51585722	16.04782862	9.54064036	50.1043262

Discussion

-limitation (smoke, risk vs actual impact, agriculture plantation consistency, crop generalization)
-policy recommendation (use to monitor, deploy resilience, potential risk in legislative responsibility conflict?)

The results of this study highlight the spatial patterns of California agriculture in relation to wildfire hazard zones, revealing which crops are most exposed to Moderate, High, and Very High fire risks. By conducting spatial and quantitative analysis, this study attempts to open up the possibility of finding concrete vulnerability of agricultural land to wildfire threats, as well as the top crops at risk. However, there are several limitations to this analysis. First, the study relies on proximity to hazard zones rather than actual fire occurrence or damage, meaning that the risk scores indicate potential exposure rather than realized impact. Thus, the extent of the practicality of this map is limited by the uncertainty in whether wildfire threats would actually materialize in those regions. Furthermore, the use of generalized crop codes may oversimplify the heterogeneity of agricultural practices, as individual fields can differ in vulnerability depending on planting schedules, irrigation, or local terrain. Moreover, temporal changes in seasonality on agriculture, as well as the potential increase/changes in FHSZ regions, could limit this study's findings usability for future assessments. These factors should be considered when interpreting the results or making operational decisions.

Additionally, wildfire impacts such as smoke exposure or secondary losses are not captured within this study, and are limited to capturing the full length of potential "exposure" of agricultural regions to the potential "impact" of wildfires. Other studies have fully focused on this topic and found that wildfire smoke contributes to elevated concentrations of fine particulate matter and other air pollutants, affecting air quality, visibility, and respiratory health for farmworkers and rural communities[7]. These findings suggest that while proximity to hazard zones provides insight into where agriculture is most *exposed* to fire risk, it does not fully account for downwind smoke distribution, air quality degradation, or health and economic effects on crops and workers. Hence, future research should further integrate these assessments to fully capture the potential exposure and proximity of this study.

Conclusion

This study analyzes the types of crops in California most prone to wildfires with the latest data. As wildfire disasters become more prominent and severe, it is important to find where and which agricultural crops are most exposed to the risk of wildfire damage. By integrating fire hazard severity maps with crop distribution data, studies can generate crop-specific risk scores, assessing which commodities are most vulnerable to varying levels of fire hazard. Such analyses not only provide a clearer picture of agricultural vulnerability but also support targeted risk mitigation strategies, inform land-use planning, and guide policy interventions aimed at protecting both the economic value and the long-term resilience of California's agricultural sector.

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[7] Voiland, A. (2025, May 28). *Dusty skies in California farm country*. **NASA Earth Observatory**. https://science.nasa.gov/earth/earth-observatory/dusty-skies-in-california-farm-country-154363/

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